

```
1  -- =====
2  -- create database
3  -- =====
4  ● CREATE DATABASE novamart;
5
6  ● use novamart;
7  -- =====
8  -- Create table customers, order, products
9  -- =====
10 ● CREATE TABLE customers (
11     Customer_ID INT PRIMARY KEY,
12     Name VARCHAR(100),
13     Email VARCHAR(100) NOT NULL,
14     Phone VARCHAR(100) NOT NULL,
15     City VARCHAR(100) NOT NULL,
16     State VARCHAR(100) DEFAULT NULL,
17     Country VARCHAR(100) NOT NULL,
18     Signup_Date DATE NOT NULL,
19     Gender VARCHAR(10) DEFAULT NULL,
20     Age INT DEFAULT NULL
21 );
22
23 ● CREATE TABLE products (
24     product_id INT PRIMARY KEY AUTO_INCREMENT,
25     Product_Name VARCHAR(100) NOT NULL,
26     Category VARCHAR(100) DEFAULT 'other',
27     Sub_Category VARCHAR(100) DEFAULT 'other',
28     Brand VARCHAR(100) DEFAULT 'other',
29     Unit_Price DECIMAL(10 , 2 ) CHECK (Unit_Price > 0.0),
30     Cost_Price DECIMAL(10 , 2 ) CHECK (Cost_Price > 0.0),
31     Stock INT DEFAULT 0
32 );
33
```

```
34 ● ○ CREATE TABLE orders (  
35     Order_ID INT PRIMARY KEY AUTO_INCREMENT,  
36     Customer_ID INT NOT NULL REFERENCES customer (customer_id),  
37     Product_ID INT NOT NULL REFERENCES customer (product_id),  
38     Order_Date DATE NOT NULL,  
39     Quantity INT NOT NULL,  
40     Unit_Price DECIMAL(10 , 2 ) CHECK (UNIT_PRICE > 0),  
41     Total_Amount DECIMAL(10 , 2 ) CHECK (Total_Amount > 0),  
42     Order_Status VARCHAR(100) DEFAULT 'Unknown',  
43     Payment_Method VARCHAR(100) DEFAULT 'Unknown'  
44 );  
45 -- Insert Data via import wizard !!!
```

```
46 -- 1. Retrieve all products belonging to the "Electronics" category.
```

```
47 • SELECT
```

```
48 *
```

```
49 FROM
```

```
50 products
```

```
51 WHERE
```

```
52 category = 'Electronics';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

product_id	Product_Name	Category	Sub_Category	Brand	Unit_Price	Cost_Price	Stock
1	NovaPhone X1 - Lite	Electronics	Smartphones	PixelForge	66.76	35.07	114
2	PixelForge Ultra - Blue	Electronics	Smartphones	QuantumGear	1024.74	607.81	44
3	Aether 12 Pro	Electronics	Smartphones	QuantumGear	73.79	38.30	258
4	Quantum S9 - Blue	Electronics	Smartphones	PixelForge	1082.54	619.37	214
5	SonicWave Max - White	Electronics	Smartphones	AetherTech	1219.85	604.70	81
6	SonicWave Max	Electronics	Smartphones	TechNova	438.46	227.42	172
7	NovaBook Air - Lite	Electronics	Laptops	TechNova	172.16	101.27	135
8	NovaBook Air - Pro	Electronics	Laptops	TechNova	705.34	358.83	193
9	ForgePro 15 - Black	Electronics	Laptops	QuantumGear	1249.21	703.35	185
10	ForgePro 15	Electronics	Laptops	SonicWave	1065.71	532.90	116
11	AetherBook Elite - Black	Electronics	Laptops	SonicWave	1303.72	699.85	232
12	Quantum Laptop 14 - Lite	Electronics	Laptops	QuantumGear	552.24	314.07	36
13	Quantum Laptop 14 - B...	Electronics	Laptops	AetherTech	281.56	161.95	83
14	SonicWave Notebook - ...	Electronics	Laptops	PixelForge	1484.59	839.39	285
15	SonicWave Notebook - ...	Electronics	Laptops	PixelForge	506.69	293.63	28
16	SonicWave Pro ANC - Pro	Electronics	Headphones	PixelForge	1213.41	654.18	33
17	AetherBuds Max - White	Electronics	Headphones	PixelForge	342.55	188.37	234

```
53 -- 2. Find all products with a Unit_Price greater than $500.
54 SELECT
55 *
56 FROM
57 products
58 WHERE
59 unit_price > 500;
```

Result Grid								
Filter Rows:								
Edit: Export/Import: Wrap Cell Content:								
	product_id	Product_Name	Category	Sub_Category	Brand	Unit_Price	Cost_Price	Stock
2	2	PixelForge Ultra - Blue	Electronics	Smartphones	QuantumGear	1024.74	607.81	44
4	4	Quantum S9 - Blue	Electronics	Smartphones	PixelForge	1082.54	619.37	214
5	5	SonicWave Max - White	Electronics	Smartphones	AetherTech	1219.85	604.70	81
8	8	NovaBook Air - Pro	Electronics	Laptops	TechNova	705.34	358.83	193
9	9	ForgePro 15 - Black	Electronics	Laptops	QuantumGear	1249.21	703.35	185
10	10	ForgePro 15	Electronics	Laptops	SonicWave	1065.71	532.90	116
11	11	AetherBook Elite - Black	Electronics	Laptops	SonicWave	1303.72	699.85	232
12	12	Quantum Laptop 14 - Lite	Electronics	Laptops	QuantumGear	552.24	314.07	36
14	14	SonicWave Notebook - Lite	Electronics	Laptops	PixelForge	1484.59	839.39	285
15	15	SonicWave Notebook - Large	Electronics	Laptops	PixelForge	506.69	293.63	28
16	16	SonicWave Pro ANC - Pro	Electronics	Headphones	PixelForge	1213.41	654.18	33
19	19	Quantum Sound X - Pro	Electronics	Headphones	PixelForge	1349.79	727.45	112
20	20	Quantum Sound X - Lite	Electronics	Headphones	PixelForge	755.46	436.76	56
21	21	NovaEars Elite - Blue	Electronics	Headphones	AetherTech	1194.35	646.67	32
22	22	ForgeAudio HD - Lite	Electronics	Headphones	TechNova	807.79	486.14	5
23	23	ForgeAudio HD - White	Electronics	Headphones	TechNova	1330.63	768.57	174
27	27	Quantum Time - Black	Electronics	Smartwatches	SonicWave	748.24	379.59	185

```
60 -- 3. List all customers from the United States.
```

```
61 SELECT
```

```
62 *
```

```
63 FROM
```

```
64     customers
```

```
65 WHERE
```

```
66     country = 'United States';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

Customer_ID	Name	Email	Phone	City	State	Country	Signup_Date	Gender	Age
3	Victoria Wyatt	jamesshaww@example.com	(555)434-1928	Lake Chad	Georgia	United States	0000-00-00	Other	37
5	Daniel Bright	ddavis@example.org	(626)291-6697	Juliechester	California	United States	0000-00-00	Female	49
6	Edward Moore	mckaynancy@example.com	728.914.8932x52	Lake Nicoleview	Florida	United States	0000-00-00	Other	26
7	Keith Miller	wcabrera@example.net	001-822-778-248	North Sarahport	Pennsylvania	United States	0000-00-00	Other	65
15	Lauren Hernandez	bobbydiaz@example.org	(833)215-8692x3	North Sierraberg	Pennsylvania	United States	0000-00-00	Other	59
16	Michael Moss	josephcobb@example.org	(707)633-7543x3	Lake Sandra	Ohio	United States	0000-00-00	Female	30
25	Michael Elliott	stevenscott@example.com	(699)812-4190	Port Andrew	New York	United States	0000-00-00	Male	52
26	Lori Gardia	melissa91@example.net	8589651850	Davidborough	New York	United States	0000-00-00	Male	60
28	Douglas Carlson	angela83@example.org	367.683.7770	Hunterborough	Florida	United States	0000-00-00	Other	38
29	Kara Schmidt	jbarajas@example.com	(574)544-3135x1	South Heather	Illinois	United States	0000-00-00	Male	26
33	Michael Bradshaw	ronaldstephens@example....	(642)910-2499	East Keithport	North Carolina	United States	0000-00-00	Male	57
34	Katherine Miller	prestonjessica@example.net	901-639-9049x02	South Haleyfurt	North Carolina	United States	0000-00-00	Female	47
39	Samantha Pollard	ihanna@example.net	001-851-471-870	West Juliefurt	New York	United States	0000-00-00	Male	26
40	Marie Christian	ofinley@example.net	001-478-609-134	South Kimberlyv...	Michigan	United States	0000-00-00	Other	63
41	Charles Schmidt	georgemiller@example.com	-4875	North Markton	Michigan	United States	0000-00-00	Other	36
43	Robert Robinson	christinebyrd@example.net	001-421-504-709	New Juleport	Texas	United States	0000-00-00	Other	59
49	Jonathan Doughe...	acostaregina@example.net	447.300.7661x77	Weststad	Illinois	United States	0000-00-00	Female	27

```
67  -- 4. Show all orders placed in November 2024.
68  SELECT
69      *
70  FROM
71      orders
72  WHERE
73      Order_Date BETWEEN '2024-11-01' AND '2024-11-30';
```

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 									
	Order_ID	Customer_ID	Product_ID	Order_Date	Quantity	Unit_Price	Total_Amount	Order_Status	Payment_Method
▶	31	285	43	2024-11-05	1	1454.41	1454.41	Completed	Google Pay
	33	259	168	2024-11-08	2	57.00	114.00	Completed	Apple Pay
	66	266	156	2024-11-15	3	62.26	186.78	Completed	Debit Card
	71	322	183	2024-11-11	1	73.25	73.25	Completed	Bank Transfer
	72	159	21	2024-11-16	1	1194.35	1194.35	Completed	PayPal
	81	117	190	2024-11-15	1	58.02	58.02	Completed	Credit Card
	93	298	25	2024-11-13	2	292.61	585.22	Shipped	Google Pay
	111	30	121	2024-11-26	3	158.60	475.80	Completed	Credit Card
	116	159	45	2024-11-05	1	33.16	33.16	Completed	Credit Card
	148	167	71	2024-11-21	4	150.45	601.80	Shipped	Google Pay
	158	160	182	2024-11-14	1	78.66	78.66	Shipped	Credit Card
	170	97	39	2024-11-19	1	949.93	949.93	Shipped	Apple Pay
	179	242	50	2024-11-10	4	1414.14	5656.56	Completed	Apple Pay
	183	392	41	2024-11-05	3	620.42	1861.26	Shipped	Apple Pay
	195	402	28	2024-11-09	2	115.14	230.28	Completed	Debit Card
	206	467	152	2024-11-01	2	332.71	665.42	Shipped	Apple Pay
	209	361	174	2024-11-23	3	131.66	394.98	Completed	Bank Transfer

orders 9 ▼

```
74 -- 5. Calculate the total stock available across all products.  
75 • SELECT  
76     SUM(stock) AS Total_Stock  
77 FROM  
78     products;  
79
```

Result Grid		 Filter Rows:	Export: 	Wrap Cell Content: 
	Total_Stock			
▶	29914			

```
--  
80  -- 6. Find the details of the most expensive product.
```

```
81  • SELECT  
82      product_name, MAX(cost_price) AS cost_price  
83  FROM  
84      products;
```

```
85  • SELECT  
86      product_name, MAX(unit_price) AS unit_price  
87  FROM  
88      products;
```

```
89
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	product_name	unit_price
▶	NovaPhone X1 - Lite	1492.86

```

90  -- 7. Show all orders where the customer ordered more than 3 quantity of a product.
91  • SELECT
92      *
93  FROM
94      orders
95  WHERE
96      quantity > 3;

```

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 									
	Order_ID	Customer_ID	Product_ID	Order_Date	Quantity	Unit_Price	Total_Amount	Order_Status	Payment_Method
▶	2	401	20	2024-08-25	5	755.46	3777.30	Completed	Google Pay
	5	70	77	2024-07-31	5	10.31	51.55	Completed	Debit Card
	8	478	145	2024-05-12	6	426.30	2557.80	Completed	Bank Transfer
	12	215	140	2023-12-14	5	162.72	813.60	Completed	Google Pay
	26	59	68	2025-02-04	5	185.89	929.45	Completed	Google Pay
	39	179	23	2024-10-08	6	1330.63	7983.78	Completed	Debit Card
	43	149	147	2024-06-27	4	400.45	1601.80	Pending	Credit Card
	44	70	193	2025-03-29	4	66.99	267.96	Shipped	Debit Card
	58	5	3	2025-02-15	6	73.79	442.74	Completed	Apple Pay
	60	196	199	2025-04-10	4	98.66	394.64	Pending	Debit Card
	65	370	156	2024-07-09	5	57.91	289.55	Completed	Debit Card
	68	185	82	2024-03-20	4	87.14	348.56	Completed	Google Pay
	69	350	45	2025-03-19	5	33.16	165.80	Completed	Google Pay
	85	115	133	2025-06-22	5	316.83	1584.15	Completed	PayPal
	89	178	163	2025-04-22	4	80.25	321.00	Completed	Bank Transfer
	92	460	134	2024-09-24	4	354.75	1419.00	Cancelled	Credit Card
	94	150	67	2025-04-27	4	136.71	546.84	Cancelled	PayPal

```

97  -- 8. Retrieve all orders where the Total_Amount exceeds $200.
98  SELECT
99      *
100 FROM
101     orders
102 WHERE
103     total_amount > 200;

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:									
	Order_ID	Customer_ID	Product_ID	Order_Date	Quantity	Unit_Price	Total_Amount	Order_Status	Payment_Method
▶	2	401	20	2024-08-25	5	755.46	3777.30	Completed	Google Pay
	3	316	51	2024-10-19	1	563.17	563.17	Shipped	Apple Pay
	6	494	102	2025-06-16	1	300.60	300.60	Completed	Google Pay
	8	478	145	2024-05-12	6	426.30	2557.80	Completed	Bank Transfer
	9	237	182	2024-05-19	3	78.66	235.98	Completed	Apple Pay
	10	345	128	2024-03-26	1	247.16	247.16	Completed	Bank Transfer
	11	217	97	2024-01-26	2	150.28	300.56	Completed	Bank Transfer
	12	215	140	2023-12-14	5	162.72	813.60	Completed	Google Pay
	13	341	91	2024-04-17	2	551.71	1103.42	Shipped	Bank Transfer
	14	332	103	2025-06-16	1	406.28	406.28	Completed	Credit Card
	15	51	67	2024-05-28	3	136.71	410.13	Pending	Google Pay
	16	114	115	2024-09-01	2	512.41	1024.82	Pending	Google Pay
	17	78	89	2025-06-29	2	537.20	1074.40	Completed	Credit Card
	19	248	90	2024-10-28	1	485.39	485.39	Completed	Credit Card
	20	154	12	2025-04-09	1	552.24	552.24	Shipped	Credit Card
	21	347	43	2025-05-31	3	1454.41	4363.23	Shipped	Debit Card
	22	170	144	2024-10-01	2	158.28	316.56	Shipped	Debit Card

```

104  -- 9. List all distinct categories and brands available in the Products table.
105  • SELECT DISTINCT
106      category
107  FROM
108      products;
109  • SELECT DISTINCT
110      brand
111  FROM
112      products;

```

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 

	Order_ID	Customer_ID	Product_ID	Order_Date	Quantity	Unit_Price	Total_Amount	Order_Status	Payment_Method
▶	2	401	20	2024-08-25	5	755.46	3777.30	Completed	Google Pay
	3	316	51	2024-10-19	1	563.17	563.17	Shipped	Apple Pay
	6	494	102	2025-06-16	1	300.60	300.60	Completed	Google Pay
	8	478	145	2024-05-12	6	426.30	2557.80	Completed	Bank Transfer
	9	237	182	2024-05-19	3	78.66	235.98	Completed	Apple Pay
	10	345	128	2024-03-26	1	247.16	247.16	Completed	Bank Transfer
	11	217	97	2024-01-26	2	150.28	300.56	Completed	Bank Transfer
	12	215	140	2023-12-14	5	162.72	813.60	Completed	Google Pay
	13	341	91	2024-04-17	2	551.71	1103.42	Shipped	Bank Transfer
	14	332	103	2025-06-16	1	406.28	406.28	Completed	Credit Card
	15	51	67	2024-05-28	3	136.71	410.13	Pending	Google Pay
	16	114	115	2024-09-01	2	512.41	1024.82	Pending	Google Pay
	17	78	89	2025-06-29	2	537.20	1074.40	Completed	Credit Card
	19	248	90	2024-10-28	1	485.39	485.39	Completed	Credit Card
	20	154	12	2025-04-09	1	552.24	552.24	Shipped	Credit Card
	21	347	43	2025-05-31	3	1454.41	4363.23	Shipped	Debit Card
	22	170	144	2024-10-01	2	158.28	316.56	Shipped	Debit Card

```
113      -- 10. Find the product with the lowest stock.
114 •    SELECT
115         product_name, MIN(stock)
116     FROM
117         products;
118
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	product_name	MIN(stock)
▶	NovaPhone X1 - Lite	4

```

119  -- 11. Calculate the total revenue generated from all Completed and Shipped orders.
120  • SELECT
121      SUM(total_amount) AS Completed_Revenue
122  FROM
123      orders
124  WHERE
125      order_status = 'Completed';
126  • SELECT
127      SUM(total_amount) AS Shipped_revenue
128  FROM
129      orders
130  WHERE
131      order_status = 'Shipped';
132

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	product_name	MIN(stock)
▶	NovaPhone X1 - Lite	4

```

133  -- =====
134  -- Section 2 Intermediate
135  -- =====
136  -- 12. Find the total quantity sold and total revenue for each product category.
137  • SELECT
138      p.Category,
139      SUM(o.Quantity) AS Totla_Qty_Sold,
140      SUM(o.Total_amount) AS total_revenue
141  FROM
142      Orders o
143      JOIN
144      products p ON o.product_id = p.product_id
145  WHERE
146      o.Order_Status IN ('completed' , 'Shiped')
147  GROUP BY p.Category
148  ORDER BY Total_revenue DESC;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	Category	Totla_Qty_Sold	total_revenue
►	Electronics	352	257306.52
	Home & Kitchen	194	79957.26
	Sports & Outdoors	225	60728.43
	Beauty & Personal Care	258	25552.28
	Clothing	208	22199.31

```
149  -- 13. Calculate the average price of products in the "Clothing" category.
150  • SELECT
151      AVG(cost_price) AS avg_price
152  FROM
153      products
154  WHERE
155      category = 'Clothing';
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	avg_price
→	47.172500

```

156  -- 14. List customers who have placed at least 3 orders.
157  •  SELECT
158      c.Name, c.Customer_id, COUNT(o.Order_ID) AS Total_Orders
159  FROM
160      customers c
161      JOIN
162      orders o ON c.Customer_Id = o.Customer_Id
163  GROUP BY c.Customer_id , c.name

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	Category	Total_Qty_Sold	total_revenue
▶	Electronics	352	257306.52
	Home & Kitchen	194	79957.26
	Sports & Outdoors	225	60728.43
	Beauty & Personal Care	258	25552.28
	Clothing	208	22199.31

```

166 -- 15. Find the most frequently ordered product (by total quantity sold).
167 • SELECT
168     p.product_id,
169     p.product_name AS Product_name,
170     SUM(o.Quantity) AS Total_Quantity_sold
171 FROM
172     orders o
173     JOIN
174     Products p ON p.product_id = o.product_id
175 WHERE
176     o.order_Status IN ('Completed' , 'Shipped')
177 GROUP BY p.Product_ID , p.product_name
178 ORDER BY Total_quantity_sold DESC
179 LIMIT 1;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	Category	Totla_Qty_Sold	total_revenue
▶	Electronics	352	257306.52
	Home & Kitchen	194	79957.26
	Sports & Outdoors	225	60728.43
	Beauty & Personal Care	258	25552.28
	Clothing	208	22199.31

```

180  -- 16. Show the Top 5 most expensive products in the Electronics category.
181  • SELECT
182      Product_Name, Brand, Unit_price
183  FROM
184      products
185  WHERE
186      Category = 'Electronics'
187  ORDER BY Unit_price DESC
188  LIMIT 5;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	Category	Total Qty Sold	total_revenue
▶	Electronics	352	257306.52
	Home & Kitchen	194	79957.26
	Sports & Outdoors	225	60728.43
	Beauty & Personal Care	258	25552.28
	Clothing	208	22199.31

```
189 -- 17. Calculate the total quantity sold by each Brand.
190 • SELECT
191     p.Brand, SUM(o.Quantity) AS Total_Quantity_sold
192 FROM
193     orders o
194     JOIN
195     products p ON p.product_id = o.product_id
196 WHERE
197     o.order_status IN ('Completed' , 'Shipped')
198 GROUP BY p.Brand;
```

Result Grid

  Filter Rows:

Export: 

Wrap Cell Content: 

	Brand	Total_Quantity_sold
▶	ActiveWear Co	45
	AdventureKit	65
	AetherTech	64
	AuraBeauty	100
	CulinaryCraft	83
	EcoCloth	24
	FitCore	66
	FreshFace	92
	GlowLab	30
	HomeEssence	65
	KitchenPro	29
	LivingSpace	28
	LuxeLine	50
	NestWell	48
	PeakPerforma...	63
	PixelForge	127
	PureEssence	24

```

199  -- 18. List the cities of customers who have spent more than $300 in total.
200  • SELECT
201      c.City, c.Name, SUM(o.Total_amount) AS Total_Amount_Spent
202  FROM
203      customers c
204  JOIN
205      orders o ON c.customer_id = o.Customer_id
206  WHERE
207      o.Order_status IN ('Completed' , 'Shipped')
208  GROUP BY c.city
209  HAVING SUM(o.total_amount) > 300
210  ORDER BY Total_Amount_Spent DESC;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	City	Name	Total_Amount_Spent
▶	North Phillip	Todd Fischer	15357.56
	Lake Natalie	Meghan Sellers	11285.94
	Port Bonniefurt	Andrew Green	10546.75
	Jenniferchester	Richard Jennings MD	8020.29
	Allenbury	Renee Melendez	7655.56
	Lisacheater	Darryl Davis	6849.08
	Tammyburgh	William Green	6514.29
	East Saramouth	Frederick Ellis II	6389.59
	Donaldport	Eric Mcclain	6049.54
	North Jacksonfort	Melissa Jackson	5971.75
	Lake Katherine	Thomas Duran	5926.68
	Michaeltown	Jennifer Smith	5501.86
	Denisebury	Keith Serrano	5412.00
	North Brenda	Sarah Roberts	5309.79
	Royfurt	James Moore	5258.23
	South Tiffanybor...	Johnathan Thomas	5183.72
	East Craigfurt	John Smith	5112.98

```

211 -- 19. Find the customer who spent the most (highest lifetime value).
212 • SELECT
213     c.Customer_ID,
214     c.Name,
215     c.Email,
216     SUM(o.Total_Amount) AS Lifetime_Value
217 FROM
218     Customers c
219     JOIN
220     Orders o ON c.Customer_ID = o.Customer_ID
221 WHERE
222     o.Order_Status IN ('Completed' , 'Shipped')
223 GROUP BY c.Customer_ID , c.Name , c.Email
224 ORDER BY Lifetime_Value DESC
225 LIMIT 1;

```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



Fetch rows:



	Customer_ID	Name	Email	Lifetime_Value
▶	79	Todd Fischer	carriemorales@example.org	15357.56

```
226 -- 20. Calculate the remaining stock of each product after fulfilling all Completed/Shipped orders.
227 • SELECT
228     p.Product_ID,
229     p.Product_Name,
230     p.Stock AS Initial_Stock,
231     COALESCE(SUM(o.Quantity), 0) AS Total_Sold,
232     (p.Stock - COALESCE(SUM(o.Quantity), 0)) AS Remaining_Stock
233 FROM
234     Products p
235     LEFT JOIN
236     Orders o ON p.Product_ID = o.Product_ID
237     AND o.Order_Status IN ('Completed' , 'Shipped')
238 GROUP BY p.Product_ID , p.Product_Name , p.Stock
239 ORDER BY Remaining_Stock ASC;
240
```

Result Grid

 Filter Rows:

Export: 

Wrap Cell Content: 

	Product_ID	Product_Name	Initial_Stock	Total_Sold	Remaining_Stock
▶	22	ForgeAudio HD - Lite	5	8	-3
	63	High-Waist Jeans	4	4	0
	173	Dry Shampoo - Blue	7	6	1
	33	NovaTab 11 - Pro	10	8	2
	109	Dining Chair Set - Black	15	8	7
	70	Denim Overalls - Black	19	10	9
	66	Cashmere Sweater - Black	20	10	10
	133	Camping Tent 4P - Pro	23	12	11
	137	Trekking Poles - Black	27	11	16
	34	AetherPad Pro	30	13	17
	188	Cologne Classic - Lite	25	7	18
	47	NovaCharge Fast - Pro	29	11	18
	16	SonicWave Pro ANC - Pro	33	14	19
	44	SonicWave Camera - Blue	30	10	20
	21	NovaEars Elite - Blue	32	12	20
	179	Eyeshadow Palette - White	28	8	20
	46	NovaCharge Fast - Lite	29	7	22

```

245  -- 21. Show the monthly revenue trend (Year-Month) for Completed and Shipped orders.
246  •  SELECT
247      DATE_FORMAT(order_date, '%Y-%m') AS year_mnth,
248      SUM(total_amount) AS monthly_revenue
249  FROM
250      orders
251  WHERE
252      order_status IN ('Completed' , 'Shipped')
253  GROUP BY DATE_FORMAT(order_date, '%Y-%m')
254  ORDER BY year_mnth DESC;
255

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	year_mnth	monthly_revenue
▶	2025-06	32543.15
	2025-05	50204.87
	2025-04	44864.52
	2025-03	39751.96
	2025-02	24286.84
	2025-01	20912.57
	2024-12	35032.83
	2024-11	37446.59
	2024-10	42385.98
	2024-09	34944.50
	2024-08	34167.83
	2024-07	33041.30
	2024-06	25582.31
	2024-05	21387.22
	2024-04	14843.08
	2024-03	27562.31
	2024-02	13822.72

```
256 -- 22. Calculate the revenue contribution percentage of each Category.
257 SELECT
258     p.category,
259     SUM(o.total_amount) AS category_rev,
260     ROUND(SUM(o.total_amount) / (SELECT
261         SUM(total_amount)
262     FROM
263         orders
264     WHERE
265         order_status IN ('completed' , 'Shipped')) * 100,
266     2) AS revenue_pct
267 FROM
268     orders o
269     JOIN
270     products p ON p.product_id = o.product_id
271 WHERE
272     o.order_status IN ('completed' , 'shipped')
273 GROUP BY p.Category;
274
```

Result Grid

 Filter Rows:

Export: 

Wrap Cell Content: 

	category	category_rev	revenue_pct
▶	Beauty & Personal Care	29926.08	5.16
	Clothing	25940.79	4.47
	Electronics	351632.40	60.61
	Home & Kitchen	102324.02	17.64
	Sports & Outdoors	70351.18	12.13

```
275 -- 23. Calculate Gross Profit and Gross Margin % for each Category. (Profit = Total_Amount - Cost_Price x Quantity)
276 • SELECT
277     p.category,
278     ROUND(SUM(o.total_amount), 2) AS Total_Revenue,
279     ROUND((SUM(o.total_amount - (p.cost_price * o.quantity))),
280           2) AS gross_profit,
281     ROUND((SUM(o.total_amount - (p.cost_price * o.quantity)) / SUM(o.total_amount)) * 100,
282           2) AS gross_profit_pct
283 FROM
284     orders o
285     JOIN
286     products p ON p.product_id = o.product_id
287 WHERE
288     o.order_status IN ('Completed' , 'Shipped')
289 GROUP BY p.category;
290
```

Result Grid

 Filter Rows:

Export: 

Wrap Cell Content: 

	category	Total_Revenue	gross_profit	gross_profit_pct
▶	Beauty & Personal Care	29926.08	19279.42	64.42
	Clothing	25940.79	15182.24	58.53
	Electronics	351632.40	150774.84	42.88
	Home & Kitchen	102324.02	49850.82	48.72
	Sports & Outdoors	70351.18	35541.24	50.52

```

291  -- 24. Find the Top 10 customers by total spend.
292  •  SELECT
293      c.name, SUM(o.total_amount) AS Total_Spent
294  FROM
295      orders o
296      JOIN
297      customers c ON c.customer_id = o.customer_id
298  WHERE
299      o.order_status IN ('Completed' , 'Shipped')
300  GROUP BY c.name
301  ORDER BY Total_Spent DESC
302  LIMIT 10;

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	category	Total_Revenue	gross_profit	gross_profit_pct
▶	Beauty & Personal Care	29926.08	19279.42	64.42
	Clothing	25940.79	15182.24	58.53
	Electronics	351632.40	150774.84	42.88
	Home & Kitchen	102324.02	49850.82	48.72
	Sports & Outdoors	70351.18	35541.24	50.52

```

303      -- 25. Calculate the cancellation rate by month.
304  •    SELECT
305          DATE_FORMAT(order_date, '%Y-%m') AS mnth,
306          COUNT(order_status) AS count_of_cancel
307  FROM
308      orders
309  WHERE
310      order_status IN ('Cancelled')
311  GROUP BY DATE_FORMAT(order_date, '%Y-%m')
312  ORDER BY mnth;
313

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	mnth	count_of_cancel			
▶	2023-12	1			
	2024-01	2			
	2024-02	2			
	2024-03	3			
	2024-04	3			
	2024-05	3			
	2024-06	4			
	2024-07	2			
	2024-08	2			
	2024-09	3			
	2024-10	4			
	2024-12	7			
	2025-01	1			
	2025-02	5			

```

314 -- 26. Find the best performing brand in each category (using ranking)
315 • WITH brand_rank AS (
316     SELECT p.category,
317     p.brand,
318     SUM(o.total_amount) AS revenue,
319     RANK() OVER(PARTITION BY p.category ORDER BY SUM(o.total_amount) DESC) AS rnk
320
321     FROM orders o
322     JOIN products p
323     ON o.product_id = p.product_id
324     WHERE order_status IN ("completed", "shipped")
325     GROUP BY p.category, p.brand
326 )
327
328 SELECT category, brand, revenue
329 FROM brand_rank
330 where rnk =1
331 ORDER BY revenue DESC;
332

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

	category	brand	revenue
▶	Electronics	PixelForge	117871.41
	Home & Kitchen	CulinaryCraft	36916.82
	Sports & Outdoors	PeakPerformance	19299.80
	Beauty & Personal Care	AuraBeauty	9526.71
	Clothing	UrbanThread	7526.32

```

333 -- 27. Segment customers into High Value / Medium Value / Low Value based on total spending
334 • WITH customer_spending AS (
335     SELECT c.name,c.customer_id,COALESCE(SUM(o.total_amount),0) AS total_spent
336     FROM customers c
337     LEFT JOIN orders o
338     ON c.customer_id = o.customer_id
339     AND o.order_status IN ("completed","shipped")
340     GROUP BY c.name,c.customer_id
341 )
342     SELECT customer_id, name , total_spent,
343     CASE
344         WHEN total_spent >= 1000 THEN 'HIGH VALUE'
345         WHEN total_spent >= 500 THEN 'Medium Value'
346         ELSE 'LOW VALUE'
347     END AS customer_segments
348     FROM customer_spending
349     ORDER BY total_spent DESC;
350

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 				
	customer_id	name	total_spent	customer_segments
▶	79	Todd Fischer	15357.56	HIGH VALUE
	242	Meghan Sellers	11285.94	HIGH VALUE
	179	Andrew Green	10546.75	HIGH VALUE
	498	Richard Jennings MD	8020.29	HIGH VALUE
	120	Darryl Davis	6849.08	HIGH VALUE
	311	William Green	6514.29	HIGH VALUE
	224	Frederick Ellis II	6389.59	HIGH VALUE
	237	Eric McDain	6049.54	HIGH VALUE
	444	Melissa Jackson	5971.75	HIGH VALUE
	238	Thomas Duran	5926.68	HIGH VALUE
	392	Jennifer Smith	5501.86	HIGH VALUE
	475	Keith Serrano	5412.00	HIGH VALUE
	416	Sarah Roberts	5309.79	HIGH VALUE
	347	James Moore	5258.23	HIGH VALUE

```

351 -- 28. Products at stock-out risk (high sales quantity + low current stock)
352 • WITH productsales AS(
353     SELECT p.product_name, p.product_id, p.stock AS current_stock, COALESCE(SUM(o.quantity), 0) as Total_sold
354     FROM products p
355     LEFT JOIN orders o
356     ON p.product_id = o.product_id
357     AND o.order_status IN ("completed", "Shipped")
358     GROUP BY p.product_name, p.product_id
359 )
360 SELECT
361     product_id,
362     product_name,
363     current_stock,
364     total_sold
365     from productsales
366     WHERE current_stock < 10
367     AND total_sold > (SELECT AVG(total_sold) FROM productsales)
368     ORDER BY current_stock ASC, Total_sold DESC;
369

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	product_id	product_name	current_stock	total_sold
▶	22	ForgeAudio HD - Lite	5	8

369
370
371
372
373
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376
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380
381
382
383

```
-- 29. Analyze performance by Payment Method (orders, average order value, and total revenue)

SELECT
    payment_method,
    ROUND(AVG(total_amount), 2) AS average_order_value,
    COUNT(order_id) AS total_order_placed,
    ROUND(SUM(total_amount), 2) AS total_revenue
FROM
    orders
WHERE
    order_status IN ('completed' , 'Shipped')
GROUP BY payment_method
ORDER BY total_revenue DESC;
```

Result Grid				
Filter Rows:		Export:		
		Wrap Cell Content:		
	payment_method	average_order_value	total_order_placed	total_revenue
▶	Google Pay	810.72	138	111879.26
	PayPal	816.11	121	98748.88
	Credit Card	845.44	114	96380.61
	Apple Pay	875.73	110	96330.81
	Bank Transfer	726.91	125	90863.55
	Debit Card	754.13	114	85971.36

```

384      -- 30. Calculate Month-over-Month revenue growth percentage.
385  • ⊖ WITH month_rev as (
386      SELECT
387          date_format(order_date, "%y-%m") as yr_mnth,
388          sum(total_amount) as revenue
389      FROM orders WHERE order_status IN ("Completed","Shipped")
390      GROUP BY yr_mnth
391  )
392
393
394      SELECT yr_mnth, revenue,
395          lag(revenue) over (ORDER BY yr_mnth) AS prev_month_rev,
396  ⊖ round(100 * (revenue - lag(revenue) OVER (ORDER BY yr_mnth))
397      /nullif(lag(revenue) OVER (ORDER BY yr_mnth),0),2) as mnt_wise_pct
398      FROM month_rev;

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 				
	yr_mnth	revenue	prev_month_rev	mnt_wise_pct
▶	23-10	14524.10	NULL	NULL
	23-11	9101.87	14524.10	-37.33
	23-12	11380.51	9101.87	25.03
	24-01	12387.41	11380.51	8.85
	24-02	13822.72	12387.41	11.59
	24-03	27562.31	13822.72	99.40
	24-04	14843.08	27562.31	-46.15
	24-05	21387.22	14843.08	44.09
	24-06	25582.31	21387.22	19.61
	24-07	33041.30	25582.31	29.16
	24-08	34167.83	33041.30	3.41
	24-09	34944.50	34167.83	2.27
	24-10	42385.98	34944.50	21.30
	24-11	37446.59	42385.98	-11.65